

Mathematical Foundations for Machine Learning - Mid-Semester 1 Shorts

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MFML Midsem 1 - Last 30 Minutes

External Sources

Intuition

Use this route in order: decide shapes and conditions, choose the smallest valid formula, calculate, then interpret. Spend 10 minutes on linear systems and geometry, 8 on decompositions, 8 on calculus, and 4 on the recall prompts.

Definitions and jargon

- Rank is the number of independent columns (equivalently, rows). Nullity is the number of free directions in $Ax=0$.
- A basis spans the space and is linearly independent. An affine space $a+U$ is a shifted subspace and need not contain zero.
- An eigenpair satisfies $Av=\lambda v$ with v nonzero. A singular value is a nonnegative scale factor in an SVD.
- A gradient collects first partial derivatives of a scalar. A Jacobian collects first derivatives of a vector. A Hessian collects second partial derivatives of a scalar.

Formulas, symbols, shapes, and assumptions

- For A of shape m by n , consistency requires $\text{rank}(A)=\text{rank}([A|b])$; uniqueness additionally requires $\text{rank}(A)=n$.
- $\text{rank}(A)+\text{nullity}(A)=n$. For orthonormal columns in Q , projection is $QQ^T x$. Only a real square orthogonal Q has $Q^{-1}=Q^T$.
- $\det(AB)=\det(A)\det(B)$ for square matrices. A real symmetric positive-definite A has $A=LL^T$.
- Every real m by n matrix has $A=U \Sigma V^T$; the number of positive singular values is $\text{rank}(A)$.
- Column-Jacobian convention: for $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, J_f is m by n , with entry $(i,j)=\partial f_i / \partial x_j$.
- $\text{grad}_x(x^T P x)=(P+P^T)x$; it is $2Px$ only when P is symmetric.
- Taylor: $f(x+h) \approx f(x)+\text{grad } f(x)^T h+(1/2)h^T H(x)h$.

Conditional properties

- A square matrix is invertible exactly when it is nonsingular, equivalently its determinant is nonzero and its rank is full.
- Distinct eigenvalues are sufficient for diagonalizability, but a zero eigenvalue still makes the matrix singular.
- At a differentiable interior local extremum, zero gradient is necessary, not sufficient. At a stationary point, positive-definite Hessian means strict local minimum, negative-definite means strict local maximum, indefinite means saddle, and semidefinite can be inconclusive. A global claim needs a global condition such as convexity.

Important distinctions

- Consistent is not the same as unique. Orthogonal is not the same as orthonormal. Diagonalizable is not the same as invertible.
- Automatic differentiation applies chain rules to recorded operations; it is neither symbolic differentiation nor finite differences. A non-scalar output needs a seed vector, producing a vector-Jacobian product in reverse mode.

Checked example

For the objects below, both coefficient and augmented matrices have rank 1, so the system is consistent. Since there are 2 unknowns and rank is 1, one free variable remains: it is not unique.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$

Traps and counterexamples

- $a+U$ may miss zero: $(1,0)+\text{span}((0,1))$ is the vertical line $x=1$.
- Distinct eigenvalues 0 and 2 make $\text{diag}(0,2)$ diagonalizable but singular.
- $f(x)=x^4$ has zero gradient and zero Hessian at zero, yet zero is a strict minimum: semidefinite evidence alone did not decide it.

Answer-free recall

- 1 State the complete rank test for no, one, and infinitely many solutions.
- 2 When is QQ^T an orthogonal projector?
- 3 State the existence condition for ordinary Cholesky.
- 4 Give the SVD rank rule and the best-rank-k rule.
- 5 State the Jacobian convention and shapes.
- 6 Classify all definite/indefinite Hessian cases at a stationary point.

Session 1 - Linear Systems and Matrices

[Know More Detailed External Sources](#)

Intuition

A linear system asks whether one list of unknown numbers can satisfy several straight-line rules at once. Writing all coefficients as a matrix turns many equations into one object that row operations can simplify.

Definitions and jargon

A is the coefficient matrix, x the unknown column vector, and b the right-hand-side column vector in $Ax=b$. The augmented matrix $[A|b]$ appends b . A pivot is a leading entry after row reduction; rank is the number of pivots. A variable without a pivot is free.

Formulas, symbols, shapes, and assumptions

If A is m by n , then x is n by 1 , b is m by 1 , and Ax is m by 1 . Matrix multiplication uses row-by-column products. Legal row operations are row swaps, multiplying a row by a nonzero number, and adding a multiple of one row to another.

Conditional properties

The system is consistent exactly when $\text{rank}(A) = \text{rank}([A|b])$. If consistent, it is unique exactly when $\text{rank}(A) = n$; otherwise it has $n - \text{rank}(A)$ free variables and infinitely many solutions. An inverse exists only for a square nonsingular matrix.

Important distinctions

Fewer pivots than unknowns does not mean inconsistency. It means non-uniqueness only after consistency has been checked. Also, $Ax=b$ has one right-hand side; $AX=B$ denotes several right-hand sides.

Checked example

Solve $x+y=5$ and $x-y=1$. Add the equations: $2x=6$, so $x=3$. Substitute into $x+y=5$: $3+y=5$, so $y=2$. There are two pivots for two unknowns, hence the solution is unique. Checking gives $3+2=5$ and $3-2=1$.

Traps and counterexamples

The equations $x+y=1$ and $2x+2y=3$ reduce to a contradiction, so $\text{rank}(A) < \text{rank}([A|b])$. The equations $x+y=1$ and $2x+2y=2$ have equal ranks but one free variable, so they have infinitely many solutions.

Answer-free recall

- 1 Give the shapes in $Ax=b$ when A is 7 by 3 .
- 2 State the rank conditions for all three solution cases.
- 3 Why must a row scaling use a nonzero number?

Session 2 - Vector Spaces, Basis, Rank, and Affine Spaces

[Know More Detailed External Sources](#)

Intuition

A vector space is a collection of objects that can be added and scaled without leaving the collection. A basis is a smallest nonredundant coordinate system for that space.

Definitions and jargon

A linear combination is $c_1 v_1 + \dots + c_k v_k$. Its span is the set of all such combinations. Vectors are independent when only all-zero coefficients produce zero. A basis is both independent and spanning. A subspace contains zero and is closed under addition and scalar multiplication. An affine space is $a+U$, a shift of subspace U by point a .

Formulas, symbols, shapes, and assumptions

For A of shape m by n , $\text{rank}(A) + \text{nullity}(A) = n$. The column space lies in $\mathbb{R}^m$, row space in $\mathbb{R}^n$, null space in $\mathbb{R}^n$, and left-null space in $\mathbb{R}^m$.

Conditional properties

More than n vectors in $\mathbb{R}^n$ must be dependent. Exactly n independent vectors in $\mathbb{R}^n$ form a basis, but fewer independent vectors need not span all of $\mathbb{R}^n$. Pivot columns of the original matrix give a column-space basis; nonzero rows of an echelon form give a row-space basis.

Important distinctions

Independent does not automatically mean a basis of the ambient space: one nonzero vector is independent in $\mathbb{R}^2$ but spans only a line. An affine space need not contain zero and is a subspace only when it shifts through zero.

Checked example

Let $v_1 = (1, 0)$ and $v_2 = (1, 1)$. If $c_1 v_1 + c_2 v_2 = (0, 0)$, the second coordinate gives $c_2 = 0$, then the first gives $c_1 = 0$; they are independent. For any (a, b) , choose $c_2 = b$ and $c_1 = a - b$; then $c_1 v_1 + c_2 v_2 = (a, b)$. They also span $\mathbb{R}^2$, so they form a basis.

Traps and counterexamples

The line $(1, 0) + \text{span}((0, 1))$ has equation $x = 1$; it does not contain $(0, 0)$, so it is affine but not a subspace. Row-reducing changes columns, so do not use columns of the reduced matrix as a basis for the original column space.

Answer-free recall

- 1 Test whether a set is a subspace.
- 2 State basis in two words joined by "and".
- 3 Locate all four fundamental subspaces for an m by n matrix.
- 4 Explain why more than n vectors in $\mathbb{R}^n$ are dependent.

Session 3 - Norms, Inner Products, Angles, and Projection

[External Sources](#)

Intuition

Norms measure size, inner products measure alignment, and projection finds the closest point in a chosen direction or subspace.

Definitions and jargon

For vector x , the Euclidean norm $\|x\|_2$ is its ordinary length. The inner product $x^T y$ is the dot product. Vectors are orthogonal when their inner product is zero and orthonormal when they are orthogonal and each has length 1.

Formulas, symbols, shapes, and assumptions

For x in $\mathbb{R}^n$, $\|x\|_1 = \sum |x_i|$, $\|x\|_2 = \sqrt{\sum x_i^2}$, and $\|x\|_\infty = \max |x_i|$. Distance is $d(x,y) = \|x-y\|$. For nonzero x,y , $\cos(\theta) = x^T y / (\|x\|_2 \|y\|_2)$. For nonzero direction u , the projection coefficient is $c = (u^T x) / (u^T u)$, the projection is cu , and the residual is $r = x - cu$. If Q has orthonormal columns, projection onto their span is $p = QQ^T x$.

Conditional properties

Every norm is nonnegative, zero only at the zero vector, scales by $|c|$, and obeys the triangle inequality. The angle formula requires both vectors nonzero. In a Gram-Schmidt step, subtract from v_j every earlier component, then normalize the residual:

$$\begin{aligned} r_j &= v_j - \sum_{i=1}^{j-1} q_i (q_i^T v_j) \\ q_j &= r_j / \|r_j\|_2, \text{ provided } r_j \neq 0 \\ r_j &= 0 \Leftrightarrow v_j \text{ is already in span}(q_1, \dots, q_{j-1}) \end{aligned}$$

If the residual is nonzero, normalization makes a new unit direction orthogonal to the earlier q_i . A zero residual means v_j was already in their span, so the inputs were dependent and no new q_j can be formed.

Important distinctions

Orthogonal vectors need not have unit length. $Q^T Q = I$ for orthonormal columns even when Q is rectangular, but $Q^{-1} = Q^T$ only for real square orthogonal Q .

Checked example

Project $x = (3,4)$ onto the direction $q = (1,0)$, which already has length 1. The coefficient is $q^T x = 3$. Thus $p = q(q^T x) = (3,0)$. The residual is $x - p = (0,4)$, and $q^T(x-p) = 0$, confirming orthogonality.

Traps and counterexamples

The zero vector has no angle because the denominator would be zero. For Q with one column $(1,0)$, $Q^T Q = 1$, but Q is 2 by 1 and has no ordinary inverse.

Answer-free recall

- 1 State the four norm axioms.
- 2 When is the angle formula defined?
- 3 Describe one Gram-Schmidt subtraction step.
- 4 State the conditions behind QQ^T and $Q^{-1} = Q^T$.

Session 4 - Determinant, Trace, Eigenpairs, and Cholesky

[Know More Detailed External Sources](#)

Intuition

The determinant tells whether a square transformation collapses space, trace summarizes its diagonal, eigenvectors keep their direction under the transformation, and Cholesky factors a well-behaved symmetric matrix.

Definitions and jargon

An eigenpair is a scalar λ and nonzero vector v satisfying $Av = \lambda v$. The trace $\text{tr}(A)$ is the sum of diagonal entries. A real matrix is symmetric when $A^T = A$; it is positive definite when $x^T A x > 0$ for every nonzero real x .

Formulas, symbols, shapes, and assumptions

For the matrix below, $\det(A) = ad - bc$. Eigenvalues solve $\det(A - \lambda I) = 0$. For square A , $\text{tr}(A)$ equals the sum of eigenvalues counted with algebraic multiplicity and $\det(A)$ equals their product. If A is m by n and B is n by m , then $\text{tr}(AB) = \text{tr}(BA)$. Ordinary Cholesky is $A = LL^T$, with lower-triangular L having positive diagonal.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Conditional properties

An ordinary real Cholesky factor exists uniquely for real symmetric positive-definite A . A square matrix is invertible exactly when its determinant is nonzero. Real symmetric matrices have real eigenvalues and an orthonormal eigenbasis.

Important distinctions

A zero eigenvalue makes a matrix singular. Distinct eigenvalues guarantee independent eigenvectors, but distinctness is sufficient rather than necessary for diagonalization.

Checked example

For the matrix below, the trace is 13, determinant is 36, and eigenpairs use the coordinate directions with eigenvalues 4 and 9. Its Cholesky factor is shown explicitly; multiplication squares its diagonal entries.

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = LL^T$$

Traps and counterexamples

Symmetry alone is insufficient for ordinary Cholesky: $\text{diag}(1, -1)$ is symmetric but not positive definite. The matrix $\text{diag}(0, 2)$ has distinct eigenvalues and is diagonalizable, yet determinant zero makes it singular.

Answer-free recall

- 1 State three equivalent nonsingularity tests.
- 2 Define an eigenpair, including the condition on v .
- 3 State Cholesky assumptions and output shape.
- 4 What special eigenvector property holds for real symmetric matrices?

Session 5 - Diagonalization, SVD, and Low-Rank Approximation

External Sources

Intuition

Decompositions replace one complicated transformation with rotations or direction changes plus simple scaling. Keeping only the strongest SVD directions compresses a matrix while controlling error.

Definitions and jargon

A square matrix is diagonalizable when it has a full eigenvector basis. The singular value decomposition (SVD) writes any real rectangular matrix as orthonormal directions, nonnegative scales, and orthonormal input directions. Rank k means at most k independent output directions.

Formulas, symbols, shapes, and assumptions

For a diagonalizable n by n matrix and any real m by n matrix respectively,

$$A = PDP^{-1}$$

$$A = U \Sigma V^T$$

Columns of P are independent eigenvectors and D contains matching eigenvalues. In the full SVD, U is m by m , Σ is m by n , and V is n by n . Singular values are ordered $\sigma_1 \geq \dots \geq \sigma_r > 0$, where $r = \text{rank}(A)$.

Conditional properties

A is diagonalizable exactly when it has n independent eigenvectors. Distinct eigenvalues are sufficient. The count of positive singular values equals rank. If A_k retains the first k singular directions, Eckart-Young gives

$$\|A - A_k\|_2 = \sigma_{k+1}$$

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_r^2}$$

The spectral norm measures the largest singular stretch; the Frobenius norm is the square root of the sum of squared singular values. σ_{k+1} is the first omitted singular value. If $k \geq r$, both errors are zero. When $k < r$ and $\sigma_k = \sigma_{k+1} > 0$, a positive tie across the keep/discard boundary preserves

the minimum error but can make the best rank- k approximation nonunique. A tie only among omitted zeros does not create distinct zero-error approximants.

Important distinctions

Every real rectangular matrix has an SVD, but eigendecomposition and diagonalization concern square matrices. Singular values are never negative; eigenvalues may be negative or complex.

Checked example

For diagonal $A = \text{diag}(3, 1)$, the SVD uses singular values $3, 1$. The best rank-1 approximation keeps 3 and replaces 1 by 0 , giving $A_1 = \text{diag}(3, 0)$. Thus

$$\begin{aligned}\|A - A_1\|_2 &= \sigma_2 = 1 \\ \|A - A_1\|_F &= \sqrt{\sigma_2^2} = 1\end{aligned}$$

Traps and counterexamples

$\text{diag}(0, 2)$ is diagonalizable but not invertible. Repeated eigenvalues do not automatically prevent diagonalization: the identity has one repeated eigenvalue and every basis is an eigenbasis.

Answer-free recall

- 1 State the exact diagonalizability condition.
- 2 Give full SVD shapes for an m by n matrix.
- 3 Relate rank to singular values.
- 4 State both Eckart-Young error formulas and the tie caveat.

Session 6 - Derivatives, Gradients, and Jacobians

External Sources

Intuition

A function is an input-output rule. A scalar function produces one number; a vector function produces an ordered list of numbers. A derivative measures local change. A gradient points toward fastest increase of a scalar output, while a Jacobian records how every output changes with every input.

Definitions and jargon

$\mathbb{R}$ means the real numbers. $\mathbb{R}^n$ means real columns with n entries, and $\mathbb{R}^m$ means real columns with m entries. We treat every input vector as a column. A superscript T means transpose: it turns a column into a row.

For a one-variable scalar rule $f: \mathbb{R} \rightarrow \mathbb{R}$, the derivative $f'(x)$ is one number: the limiting slope at input x . The partial derivative $\partial f / \partial x_i$ changes only input x_i while holding the other inputs fixed. For scalar $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient $\text{grad } f$ is the n by 1 column of partial derivatives. We use the output-by-input Jacobian convention: for vector function $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$, J_g is m by n , and entry (i, j) is $\partial g_i / \partial x_j$.

Formulas, symbols, shapes, and assumptions

Sum, product, and quotient rules apply as in one variable. In the one-variable chain rule, x , $y=g(x)$, and $z=f(y)$ are scalars, and the derivative factors multiply in path order.

For the multivariable chain rule, let column x be n by 1 , let $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$, and let $f: \mathbb{R}^m \rightarrow \mathbb{R}^p$. Then J_g is m by n , J_f is p by m , and the composed Jacobian is p by n . The order is outer Jacobian times inner Jacobian; reversing it is generally invalid. For a small n by 1 displacement h , the same shapes make the scalar and vector linearizations below valid.

$$z=f(y), y=g(x): dz/dx = (df/dy)(dy/dx)$$

$$J_{(f \circ g)}(x) = J_f(g(x)) J_g(x): p \text{ by } n = (p \text{ by } m)(m \text{ by } n)$$

$$f(x+h) \approx f(x) + \text{grad } f(x)^T h$$

$$g(x+h) \approx g(x) + J_g(x)h$$

Conditional properties

These linear approximations describe local behavior; their error becomes small relative to $\|h\|$ as h tends to zero. Mixed-partial equality needs suitable continuity assumptions, not merely the existence of derivatives.

Important distinctions

The gradient belongs to a scalar-output function; the Jacobian handles several outputs. Under the declared convention, a scalar function's Jacobian is 1 by n , while its gradient is the transpose-shaped n by 1 column.

Checked example

Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be $g(x,y)=(x^2+y, xy)$. The 2 by 1 input column has entries x,y . The first output has derivatives $(2x,1)$ and the second (y,x) . At $(1,2)$, the 2 by 2 Jacobian has rows $(2,1)$ and $(2,1)$. For the 2 by 1 displacement $h=(0.01,0)$, matrix-vector multiplication predicts the 2 by 1 output change $(0.02,0.02)$.

Traps and counterexamples

Do not add partial derivatives to form a gradient; keep their ordered vector shape. Do not transpose a Jacobian silently: it changes which multiplication is valid.

Answer-free recall

- 1 State both chain rules with their multiplication order and shapes.
- 2 Give gradient and Jacobian shapes for three inputs and two outputs.
- 3 Write the first-order linearization for scalar and vector outputs.

Session 7 - Matrix Gradients, Backpropagation, and Autodiff

[External Sources](#)

Intuition

Backpropagation applies the chain rule backward through a computation graph so that shared intermediate results are reused. Automatic differentiation (AD) performs this process on recorded elementary operations.

Definitions and jargon

A computation graph connects inputs, intermediate values, and outputs. A local derivative belongs to one edge. Reverse-mode AD propagates an upstream seed or cotangent backward. For scalar loss L , the usual seed is 1.

Formulas, symbols, shapes, and assumptions

For column x and constant compatible objects,

$$\begin{aligned}\text{grad}_x(a^T x) &= a \\ \text{grad}_x(x^T x) &= 2x \\ \text{grad}_x(x^T P x) &= (P + P^T)x\end{aligned}$$

The shortcut $2Px$ requires symmetric P . If output y is non-scalar and seed column v matches y , reverse mode computes the input-sensitivity column below, a vector-Jacobian product under this convention.

$$\text{input sensitivity} = Jy^T v$$

Conditional properties

Reverse mode is especially efficient for one scalar loss and many parameters. Forward mode is often attractive for few inputs and many outputs. Nondifferentiable points require a documented software convention or subgradient rule.

Important distinctions

AD is not symbolic algebra: it evaluates derivatives of executed operations at supplied values. It is not finite differences: it does not estimate slopes by perturbing inputs.

Checked example

Let $u=x^2$, $y=2u+1$, and $L=y^2$ at $x=1$. Forward values are $u=1$, $y=3$, $L=9$. Backward: $dL/dy=2y=6$, $dy/du=2$, $du/dx=2x=2$. Multiplication gives $dL/dx=6*2*2=24$.

Traps and counterexamples

For nonsymmetric P , writing $2Px$ is wrong. For example, the quadratic depends only on the symmetric part of P . Calling backward on a vector output without a matching seed leaves the requested scalar combination unspecified.

Answer-free recall

- 1 Reproduce a forward and backward pass on a three-operation graph.

- 2 State the general quadratic-form gradient and its shortcut condition.
- 3 Contrast AD with symbolic differentiation and finite differences.
- 4 Explain the non-scalar seed and its shape.

Session 8 - Hessians, Taylor Models, and Unconstrained Extrema

[External Sources](#)

Intuition

The gradient gives slope; the Hessian describes how slope changes. Together they form local Taylor models that can classify curved stationary points.

Definitions and jargon

For scalar $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian H is the n by n matrix with entry $(i,j) = \partial^2 f / (\partial x_i \partial x_j)$. A stationary point has zero gradient. A saddle rises in some directions and falls in others.

Formulas, symbols, shapes, and assumptions

With h an n by 1 displacement,

$$\text{first order: } f(x+h) \approx f(x) + \text{grad } f(x)^T h$$

$$\text{second order: } f(x+h) \approx f(x) + \text{grad } f(x)^T h + (1/2) h^T H(x) h$$

These are local approximations requiring suitable differentiability near x .

Conditional properties

At a differentiable interior local extremum, stationarity is necessary. At a stationary point, positive-definite H gives a strict local minimum, negative-definite gives a strict local maximum, and indefinite gives a saddle. Positive- or negative-semidefinite can be inconclusive. Convexity over the whole domain can promote a local minimum to a global one.

Important distinctions

Stationary does not mean extremum. A local statement concerns a neighborhood; a global statement compares every point in the domain. Linearization uses only first derivatives; the quadratic Taylor model also uses curvature.

Checked example

For $f(x,y) = x^2 + 3y^2 - 2x + 6y$, the gradient is $(2x-2, 6y+6)$. Solving gives $(1, -1)$. The Hessian is diagonal with positive entries 2, 6, so it is positive definite. Completing squares gives $f = (x-1)^2 + 3(y+1)^2 - 4$, independently confirming the unique global minimum value -4 .

Traps and counterexamples

At zero, $f(x)=x^3$ has zero derivative and zero second derivative but is not an extremum. For $f(x,y)=x^2-y^2$, the Hessian is indefinite and the origin is a saddle.

Answer-free recall

- 1 Write first- and second-order multivariate Taylor models with shapes.
- 2 State all Hessian classification cases and the semidefinite caveat.
- 3 What conditions make stationarity necessary?
- 4 What extra reasoning supports a global minimum claim?